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Semiconductor device with stress relief feature and method therefor

Abstract

A method of manufacturing a semiconductor device is provided. The method includes placing a package substrate on a carrier substrate, forming a frame on the package substrate, and affixing an active side of a semiconductor die on the package substrate. The semiconductor die together with the frame and the package substrate form a cavity between the semiconductor die and the package substrate. At least a portion of the semiconductor die and the package substrate are encapsulated with an encapsulant. The frame is configured to prevent the encapsulant from entering the cavity.

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Background/Summary

BACKGROUND

Field

(1) This disclosure relates generally to semiconductor device packaging, and more specifically, to semiconductor devices with a stress relief feature and method of forming the same.

Related Art

(2) Today, there is an increasing trend to include sophisticated semiconductor devices in products and systems that are used every day. These sophisticated semiconductor devices may include features for specific applications which may impact the configuration of the semiconductor device packages, for example. For some features and applications, the configuration of the semiconductor device packages may be susceptible to lower reliability, lower performance, and higher product or system costs. Accordingly, significant challenges exist in accommodating these features and applications while minimizing the impact on semiconductor devices' reliability, performance, and costs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention is illustrated by way of example and is not limited by the accompanying figures, in which like references indicate similar elements. Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale.

(2) FIG. 1 illustrates, in a simplified bottom-side-up plan view, an example semiconductor device having a stress relief feature at a stage of manufacture in accordance with an embodiment.

(3) FIG. 2 through FIG. 12 illustrate, in simplified cross-sectional views, the example semiconductor device at stages of manufacture in accordance with an embodiment.

DETAILED DESCRIPTION

(4) Generally, there is provided, a semiconductor device with a stress relief feature. The stress relief feature of the semiconductor device includes a cavity frame formed on a package substrate. A semiconductor die is affixed over the frame and attached to the package substrate to form a cavity between the semiconductor die and the package substrate. The frame acts a barrier to prevent mold compound from entering the cavity during an encapsulation operation. Because the cavity remains void of encapsulant material, the cavity serves as a stress relief feature. The cavity is located adjacent to an active region of the semiconductor die such that a stress sensitive circuit or sensor of the active region is substantially surrounded by the cavity frame. By forming the stress relief feature in this manner, stress sensitive circuits or sensors are not affected by the physical stresses of the encapsulant.

(5) FIG. 1 illustrates, in a simplified bottom-side-up plan view, an example semiconductor device **100** having a stress relief feature at a stage of manufacture in accordance with an embodiment. In this embodiment, a bottom side of an interconnecting package substrate **104** applied on the semiconductor device **100** is depicted. A plurality of conductive connectors **106** (e.g., solder balls) are affixed to the bottom side of the package substrate **104**. An underlying semiconductor die **102** (shown as dashed outline for reference) are embedded in an encapsulant (not shown) of the semiconductor device **100**. The number and arrangement of the conductive connectors **106** in this embodiment are chosen for illustration purposes. The term “conductive,” as used herein, generally refers to electrical conductivity unless otherwise specified. Simplified cross-sectional views of the example semiconductor device **100** taken along line A-A at stages of manufacture are depicted in FIG. 2 through FIG. 12.

(6) FIG. 2 illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a stage of manufacture in accordance with an embodiment. At this stage, the semiconductor device **100** includes a first package substrate **218** placed on a carrier substrate **202**. The carrier substrate **202** includes a releasable adhesive **204** applied on a top surface. The carrier substrate **202** is configured and arranged to provide a temporary structure for placement of the package substrate **218** and subsequent steps at stages of manufacture, for example. In this embodiment, the package substrate **218** corresponds to the package substrate **104** shown in FIG. 1.

(7) The package substrate **218** includes conductive features (e.g., traces **208-216**) surrounded by non-conductive material **206** (e.g., dielectric). In this embodiment, the conductive traces **208-216** are formed from patterned metal (e.g., copper) layers separated by dielectric layers of the package substrate **218**. The traces **208-216** are formed in the package substrate to interconnect a semiconductor die with a printed circuit board (PCB), for example. In this embodiment, portions of the traces **208** and **210** are exposed at the top side of the package substrate **218** and configured to serve as conductive connection pads for connection of a semiconductor die and through package vias (TPVs) at subsequent stages of manufacture. Portions of the traces **212** and **214** are configured to serve as conductive substrate pads at the bottom side of the package substrate **218** for attachment of conductive connectors at a subsequent stage of manufacture. The package substrate **218** may be characterized as a redistribution layer substrate and may be provided as a pre-formed package substrate placed on the carrier substrate **202** or may be formed as a build-up package substrate directly on the carrier substrate **202**.

(8) FIG. 3 illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, an opening **302** is formed through the package substrate **218**. The opening **302** may be formed before the package substrate **218** is placed on the carrier substrate **202**. In this embodiment, the opening **302** through the package substrate **218** may be configured as a passageway to allow pressure equalization with an embedded cavity formed at a subsequent stage of manufacture.

(9) FIG. 4 illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, a cavity frame **402** is placed, dispensed, or otherwise formed on the package substrate **218**. The frame **402** is configured to form a cavity at a predetermined location when a semiconductor die is affixed over the frame at a subsequent stage of manufacture, for example. In this embodiment, the frame **402** is formed as an enclosed ring having a predetermined wall width dimension **404** (e.g., ~10 microns or less) and wall height dimension **406** (e.g., ~10-50 microns). The wall dimensions **404** and **406** may be chosen to sufficiently prevent mold compound from breaching the frame **402** during an encapsulation operation at a subsequent stage, for example. The frame **402** may be formed having any suitable shape such as orthogonal shapes (e.g., square, rectangular), circular, oval, and combinations thereof. The frame **402** may be formed from suitable materials such as gel coat, epoxy, silicone, and the like to prevent mold compound from entering a cavity formed in part by the frame **402**.

(10) FIG. 5 illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, a first semiconductor die **502** is affixed to the package substrate **218**. In this embodiment, the semiconductor die **502** corresponds to the underlying semiconductor die **102** depicted in FIG. 1. The semiconductor die **502** is positioned over the frame **402** and in direct contact with the frame when affixed to the package substrate **218** by way of conductive die connectors **508**. With the semiconductor die **502** attached to the package substrate **218** and in contact with the frame **402**, a cavity **510** is formed. The opening **302** through the package substrate **218** may be configured as a passageway to allow pressure equalization between the cavity **510** and the atmosphere immediately surrounding the semiconductor device **100**, for example.

(11) The semiconductor die **502** has an active side (e.g., major side having circuitry) and a backside

(e.g., major side opposite of the active side). The semiconductor die **502** includes an active region **504** (e.g., circuitry, bond pads, etc.) formed at the active side. Bond pads (not shown) of the active region **504** may be configured for connection to respective exposed portions of the traces **210** serving as conductive connection pads by way of conductive die connectors **508**, for example. In this embodiment, semiconductor die **502** is configured in an active-side-down orientation with the active side affixed to the package substrate **218**. The cavity **510** is formed between the active side of the semiconductor die **502** and the package substrate **218** with the walls of the frame **402** substantially surrounding at least a portion of the active region **504**. The portion of the active region **504** adjacent to the cavity **510** may include features such as circuitry and/or sensors which are characterized as stress sensitive features.

(12) The semiconductor die **502** may be formed from any suitable semiconductor material, such as silicon, germanium, gallium arsenide, gallium nitride, and the like. The semiconductor die **502** may further include digital circuitry, analog circuitry, RF circuitry, a memory, a processor, a sensor, the like, and combinations thereof at the active side.

(13) In this embodiment, the conductive die connectors **508** have a first end affixed to the active region **504** of the semiconductor die **502** and a second end affixed to respective exposed portions of the traces **210** to form electrically conductive interconnects between the semiconductor die **502** and the package substrate **218**. The conductive die connectors **508** may be in the form of any suitable conductive structures such as gold studs, copper pillars, solder balls, and the like. In some embodiments, the semiconductor die **502** may further include one or more conductive through silicon vias (TSVs) **506**. Accordingly, one or more of the TSVs **506** may be interconnected with conductive traces (e.g., trace **210**) of the package substrate **218** by way of conductive die connectors **508**. The location and number of the TSVs **506** are chosen for illustration purposes.

(14) FIG. **6** illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, a second semiconductor die **602** is affixed to the first semiconductor die **502** in a stacked die configuration. The semiconductor die **602** is affixed to the semiconductor die **502** by way of conductive die connectors **608**.

(15) The semiconductor die **602** has an active side (e.g., major side having circuitry) and a backside (e.g., major side opposite of the active side). The semiconductor die **602** includes an active region **604** (e.g., circuitry, bond pads, etc.) formed at the active side. Bond pads (not shown) of the active region **604** may be configured for connection to respective TSVs **506** of the semiconductor die **502** by way of conductive die connectors **608**, for example. In this embodiment, semiconductor die **602** is configured in an active-side-down orientation with the active side affixed to the backside (e.g., TSVs **506**) of the semiconductor die **502**. The semiconductor die **602** may be formed from any suitable semiconductor material, such as silicon, germanium, gallium arsenide, gallium nitride, and the like. The semiconductor die **602** may further include digital circuitry, analog circuitry, RF circuitry, a memory, a processor, a sensor, the like, and combinations thereof at the active side.

(16) In this embodiment, the conductive die connectors **608** have a first end affixed to the active region **604** of the semiconductor die **602** and a second end affixed to respective TSVs **506** to form electrically conductive interconnects between the semiconductor die **602** and the semiconductor die **502** and the package substrate **218**. The conductive die connectors **608** may be in the form of any suitable conductive structures such as gold studs, copper pillars, solder balls, and the like. In some embodiments, the semiconductor die **602** may further include one or more through silicon vias (TSVs) **606**. For example, one or more of the TSVs **606** may be interconnected with the semiconductor die **502** by way of conductive die connectors **608** and TSVs **506** and may be further interconnected to the conductive traces (e.g., trace **210**) of the package substrate **218** by way of conductive die connectors **508**. The location and number of the TSVs **606** are chosen for illustration purposes.

(17) FIG. **7** illustrates, in a simplified cross-sectional view, the example semiconductor device **100**

taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, the semiconductor device **100** includes the stacked die (e.g., semiconductor die **502** and **602**) and the package substrate **218** at least partially encapsulated with an encapsulant **702** (e.g., epoxy molding compound) while temporarily affixed on the carrier substrate **202**. In this embodiment, the semiconductor die **502** and **602** and the package substrate **218** are over-molded with the encapsulant **702** by way of a molding process. The frame **402** serves as a barrier prohibiting the encapsulant **702** from entering the cavity **510**. Because the cavity **510** is void of encapsulant material, the encapsulant **702** is prevented from interacting with the portion of the active region **504** adjacent to the cavity **510**. Accordingly, the portion of the active region **504** adjacent to the cavity **510** may include stress sensitive features without being affected by physical stresses of the encapsulant.

(18) FIG. **8** illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, the backside of the semiconductor die **602** and portions of the TSVs **606** are revealed at the surface **802** of the encapsulant **702**. In this embodiment, the encapsulated semiconductor device **100** is subjected to a grind operation to expose the backside of the semiconductor die **602** and portions of the TSVs **606**. Alternatively, the semiconductor die **502** and **602** and the package substrate **218** may be molded with the encapsulant **702** by way of a film-assisted molding (FAM) process. For example, a FAM tool may be engaged with the backside of the semiconductor die **602** during the FAM molding process to keep the backside of the semiconductor die **602** and portions of the TSVs **606** free from encapsulant. In this manner, the backside of the semiconductor die **602** and portions of the TSVs **606** may be exposed at the surface **802** of the encapsulant **702** without a grind operation.

(19) FIG. **9** illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, one or more conductive through package vias (TPVs) **902** may be formed through the encapsulant **702**. In this embodiment, the one or more of the TPVs **902** have a first end exposed at the surface **802** of the encapsulant **702** and a second end electrically connected to conductive traces (e.g., trace **208**) of the package substrate **218**. The location and number of the TPVs **902** are chosen for illustration purposes.

(20) FIG. **10** illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, the semiconductor device **100** includes a second package substrate **1014** applied on the backside of the semiconductor die **602** and the surface **802** of the encapsulant **702**. The package substrate **1014** may be characterized as a redistribution layer substrate and may be provided as a pre-formed package substrate affixed on the backside of the semiconductor die **602** and the surface **802** of the encapsulant **702** or may be formed as a build-up package substrate directly on the on the backside of the semiconductor die **602** and the surface **802**.

(21) The package substrate **1014** includes conductive features (e.g., traces **1004-1012**) surrounded by non-conductive material **1002** (e.g., dielectric). In this embodiment, the conductive traces **1004-1012** are formed from patterned metal (e.g., copper) layers separated by dielectric layers of the package substrate **1014**. The traces **1004-1012** are formed in the package substrate **1014** to interconnect the semiconductor device **100** with a PCB or external device, for example. In this embodiment, portions of the traces **1006** are conductively connected to the TSVs **606** exposed at the backside of the semiconductor die **602** and portions of the traces **1004** are conductively connected to the TPVs **902** exposed at the surface **802** of the encapsulant **702**. Portions of the traces **1008** and **1010** are configured to serve as conductive substrate pads for attachment of conductive connectors at a subsequent stage of manufacture.

(22) FIG. **11** illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At

this stage, the semiconductor device **100** includes a plurality of conductive connectors **1102** (e.g., solder balls) affixed to the package substrate **1014**. In this embodiment, openings are formed in the non-conductive material **1002** to expose portions of the conductive traces **1008** and **1010** and the conductive connectors **1102** are attached to the exposed portions of the traces **1008** and **1010**. The conductive connectors **1102** are configured and arranged to provide conductive connections between the semiconductor device **100** and a PCB or external device, for example. The conductive connectors **1102** may be in the form of any suitable conductive structures such as solder balls, gold studs, copper pillars, and the like, to connect conductive features of the semiconductor device **100** with the PCB or external device. As an alternative to attaching conductive connectors **1102**, the exposed portions of the traces **1008** and **1010** may be plated for subsequent connection with the PCB or external device by way of solder paste attachment or other suitable conductive attachment processes (e.g., ACF, ACP).

(23) FIG. 12 illustrates, in a simplified cross-sectional view, the example semiconductor device **100** taken along line A-A at a subsequent stage of manufacture in accordance with an embodiment. At this stage, the semiconductor device **100** is separated from the carrier substrate **202** and a plurality of conductive connectors **1202** (e.g., solder balls) are affixed to the package substrate **218**. In this embodiment, openings are formed in the non-conductive material **206** to expose portions of the conductive traces **212** and **214** and the conductive connectors **1202** are attached to the exposed portions of the traces **212** and **214**. The conductive connectors **1202** are configured and arranged to provide conductive connections between the semiconductor device **100** and a PCB or external device, for example. The conductive connectors **1202** may be in the form of any suitable conductive structures such as solder balls, gold studs, copper pillars, and the like, to connect conductive features of the semiconductor device **100** with the PCB or external device. As an alternative to attaching conductive connectors **1202**, the exposed portions of the traces **212** and **214** may be plated for subsequent connection with the PCB or external device by way of solder paste attachment or other suitable conductive attachment processes (e.g., ACF, ACP).

(24) Generally, there is provided, a method of manufacturing a semiconductor device including placing a first package substrate on a carrier substrate; forming a frame on the first package substrate; affixing an active side of a first semiconductor die on the package substrate, the first semiconductor die together with the frame and the first package substrate forming a cavity between the first semiconductor die and the first package substrate; and encapsulating with an encapsulant at least a portion of the first semiconductor die and the first package substrate, the frame preventing the encapsulant from entering the cavity. The active side of the first semiconductor die may be interconnected to a conductive trace of the first package substrate by way of a conductive die connector. The frame may be configured to substantially surround a portion of the active side of the first semiconductor die, the portion of the active side including circuitry characterized as stress sensitive circuitry. The package substrate may include an opening through the first package substrate, the opening configured to allow pressure equalization between the cavity and atmosphere immediately surrounding the semiconductor device. The first semiconductor die may include a through silicon via (TSV), the TSV forming an electrically conductive interconnect between the active side and a backside of the first semiconductor die. The method may further include affixing an active side of a second semiconductor die on the backside of the first semiconductor die before encapsulating with an encapsulant, the second semiconductor die electrically interconnected with the first semiconductor die by way of the TSV. The method may further include forming a through package via (TPV), the TPV forming an electrically conductive interconnect between a first major side of the encapsulant adjacent to the first package substrate and a second major side of the encapsulant. The method may further include applying a second package substrate on the second major side of the encapsulant. The method may further include interconnecting a conductive trace of the second package substrate with a conductive trace of the first package substrate by way of the TPV.

(25) In another embodiment, there is provided, a semiconductor device including a frame formed on a first package substrate; an active side of a first semiconductor die affixed on the first package substrate, the first semiconductor die together with the frame and the first package substrate forming a cavity between the first semiconductor die and the first package substrate; and an encapsulant encapsulating at least a portion of the first semiconductor die and the first package substrate, the cavity void of the encapsulant. The active side of the first semiconductor die may be interconnected to a conductive trace of the first package substrate by way of a conductive die connector. The frame may substantially surround a portion of the active side of the first semiconductor die, the portion of the active side including circuitry characterized as stress sensitive circuitry. The package substrate may include an opening through the first package substrate, the opening configured to allow pressure equalization between the cavity and atmosphere immediately surrounding the semiconductor device. The first semiconductor die may include a through silicon via (TSV), the TSV forming an electrically conductive interconnect between the active side and a backside of the first semiconductor die. The device may further include affixing an active side of a second semiconductor die on the backside of the first semiconductor die, the second semiconductor die electrically interconnected with the first semiconductor die by way of the TSV.

(26) In yet another embodiment, there is provided, a method including placing a first package substrate on a carrier substrate; forming a frame on the first package substrate; affixing an active side of a first semiconductor die on the package substrate, the first semiconductor die located over the frame forming a cavity between the first semiconductor die and the first package substrate; and encapsulating with an encapsulant at least a portion of the first semiconductor die and the first package substrate, the frame preventing the encapsulant from entering the cavity. The active side of the first semiconductor die may be interconnected to a conductive trace of the first package substrate by way of a conductive die connector. The frame may be configured to substantially surround a portion of the active side of the first semiconductor die, the portion of the active side including circuitry characterized as stress sensitive circuitry. The package substrate may include an opening through the first package substrate, the opening configured to allow pressure equalization between the cavity and atmosphere immediately surrounding the semiconductor device. The method may further include applying a second package substrate over a major side of the encapsulant opposite of the first package substrate, the second package substrate interconnected with the first package substrate by way of a through package via (TPV).

(27) By now, it should be appreciated that there has been provided a semiconductor device with a stress relief feature. The stress relief feature of the semiconductor device includes a cavity frame formed on a package substrate. A semiconductor die is affixed over the frame and attached to the package substrate to form a cavity between the semiconductor die and the package substrate. The frame acts a barrier to prevent mold compound from entering the cavity during an encapsulation operation. Because the cavity remains void of encapsulant material, the cavity serves as a stress relief feature. The cavity is located adjacent to an active region of the semiconductor die such that a stress sensitive circuit or sensor of the active region is substantially surrounded by the cavity frame. By forming the stress relief feature in this manner, stress sensitive circuits or sensors are not affected by the physical stresses of the encapsulant.

(28) The terms “front,” “back,” “top,” “bottom,” “over,” “under” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the invention described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

(29) Although the invention is described herein with reference to specific embodiments, various modifications and changes can be made without departing from the scope of the present invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included

within the scope of the present invention. Any benefits, advantages, or solutions to problems that are described herein with regard to specific embodiments are not intended to be construed as a critical, required, or essential feature or element of any or all the claims.

(30) Furthermore, the terms “a” or “an,” as used herein, are defined as one or more than one. Also, the use of introductory phrases such as “at least one” and “one or more” in the claims should not be construed to imply that the introduction of another claim element by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim element to inventions containing only one such element, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an.” The same holds true for the use of definite articles.

(31) Unless stated otherwise, terms such as “first” and “second” are used to arbitrarily distinguish between the elements such terms describe. Thus, these terms are not necessarily intended to indicate temporal or other prioritization of such elements.

Claims

1. A method of manufacturing a semiconductor device, the method comprising: placing a first package substrate on a carrier substrate; forming a frame on the first package substrate; affixing an active side of a first semiconductor die on the package substrate, the first semiconductor die together with the frame and the first package substrate forming a cavity between the first semiconductor die and the first package substrate; and encapsulating with an encapsulant at least a portion of the first semiconductor die and the first package substrate, the frame preventing the encapsulant from entering the cavity.
2. The method of claim 1, wherein the active side of the first semiconductor die is interconnected to a conductive trace of the first package substrate by way of a conductive die connector.
3. The method of claim 1, wherein the frame is configured to substantially surround a portion of the active side of the first semiconductor die, the portion of the active side including circuitry characterized as stress sensitive circuitry.
4. The method of claim 1, wherein the package substrate includes an opening through the first package substrate, the opening configured to allow pressure equalization between the cavity and atmosphere immediately surrounding the semiconductor device.
5. The method of claim 1, wherein the first semiconductor die includes a through silicon via (TSV), the TSV forming an electrically conductive interconnect between the active side and a backside of the first semiconductor die.
6. The method of claim 5, further comprising affixing an active side of a second semiconductor die on the backside of the first semiconductor die before encapsulating with an encapsulant, the second semiconductor die electrically interconnected with the first semiconductor die by way of the TSV.
7. The method of claim 1, further comprising forming a through package via (TPV), the TPV forming an electrically conductive interconnect between a first major side of the encapsulant adjacent to the first package substrate and a second major side of the encapsulant.
8. The method of claim 7, further comprising applying a second package substrate on the second major side of the encapsulant.
9. The method of claim 8, further comprising interconnecting a conductive trace of the second package substrate with a conductive trace of the first package substrate by way of the TPV.
10. A method comprising: placing a first package substrate on a carrier substrate; forming a frame on the first package substrate; affixing an active side of a first semiconductor die on the package substrate, the first semiconductor die located over the frame forming a cavity between the first semiconductor die and the first package substrate; and encapsulating with an encapsulant at least a portion of the first semiconductor die and the first package substrate, the frame preventing the encapsulant from entering the cavity.

11. The method of claim 10, wherein the active side of the first semiconductor die is interconnected to a conductive trace of the first package substrate by way of a conductive die connector.
 12. The method of claim 10, wherein the frame is configured to substantially surround a portion of the active side of the first semiconductor die, the portion of the active side including circuitry characterized as stress sensitive circuitry.
 13. The method of claim 10, wherein the package substrate includes an opening through the first package substrate, the opening configured to allow pressure equalization between the cavity and atmosphere immediately surrounding the semiconductor device.
 14. The method of claim 10, further comprising applying a second package substrate over a major side of the encapsulant opposite of the first package substrate, the second package substrate interconnected with the first package substrate by way of a through package via (TPV).
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